0029

National University of Computer and Emerging Sciences

Feedback Control Systems (EE3004)

Date: June 6th 2024

Course Instructor

Dr. Omer Saleem (BEE-6A)

Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 05

Semester: SP-2024

Campus: Lahore

Dept: Electrical Engg.

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Student Name ___ Roll No ___ Section ___ Student Signature

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CLO # 1: Develop the mathematical model of a physical system.

Q1: **[10 + 10 marks]**

a. Find the state space representation of the following mechanical system.

$X(s)(Ms^2 + F_{v}s + k) = F(s).$

$\cancel{M\ddot{z}(t)}$

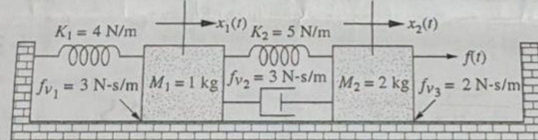
$M\ddot{x}(t) + F_v\dot{x}(t) + k$

$z_1 = \dot{x}(t)$

$z_2 = \dot{x}(t) = \dot{z}_1$

$\dot{z}_2 = \ddot{x}(t) = \ddot{z}_1$

$(sI - A)^{-1}$



b. Convert the following state space representation into a transfer function of the form, Y(s)/U(s). Hence, find the expression of output, $y(t)$, in time-domain if the input signal, $x(t)$, is a unit-step signal.

$$\frac{Y(s)}{U(s)} = \frac{4+s}{s^2+5s+4}$$

$$\dot{x}(t) = \overset{A}{\begin{bmatrix} 0 & 2 \\ -2 & -5 \end{bmatrix}} x(t) + \overset{B}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}} u(t)$$

$$y(t) = \overset{C}{\begin{bmatrix} 2 & 1 \end{bmatrix}} x(t) + \overset{D}{[0]} u(t)$$

$C(sI - A)^{-1}B + D$ $\quad \frac{y(t)}{x(t)}$

CLO # 2: Analyze the transient and steady state response of a feedback system.

Q2: The system shown below has its transient response altered by adding a tachometer. **[20 marks]**

a. Reduce the block diagram to find the system's open-loop transfer function **G(s)** as well as overall transfer function **T(s)**.

b. Design K and K_2 in the system to yield a damping ratio of **0.69** and a natural frequency of **10.0 rad/s**.

c. What is the steady-state error of this closed-loop system.

d. What modification is needed in the pre-amplifier K to ensure the steady-state error becomes zero.

$T(s) = \quad s+1$

$s^0(s^2 + 13.8s + 100.)$

Type 0 $\to \frac{1}{1+k_p}$

$s^2 + 13.8s$

$s(s + 13$

range?

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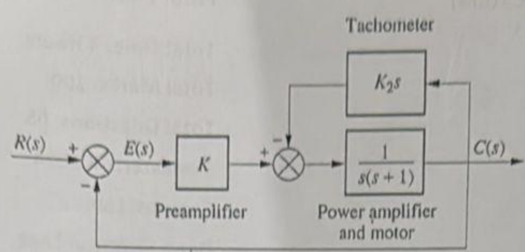


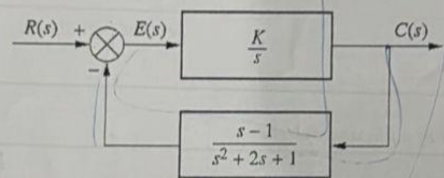
CLO # 3: Analyze the stability of a system.

Q3: **[20 marks]**

a. For the unity feedback system shown below:

i. Find the closed-loop transfer function, **T(s)**, of the system.

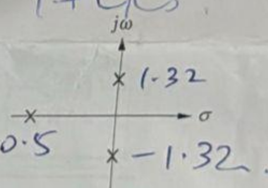
ii. Hence, use **T(s)** to find the range of ***K*** for stability.



$= \frac{G(s)}{1+G(s)H(s)}$

$\frac{G(s)}{1+G(s)}$

b. Use Routh Hurwitz criteria to find the value of ***K*** that will place the closed-loop poles of the following unity feedback system as shown in the Figure here; one pole on real-axis and two poles on $j\omega$-axis. Find the value of the poles as well.



$$G(s) = \frac{Ks^2(s+1)}{s^3+s^2+2s+K}$$

$-\frac{2}{3} < K < 0$

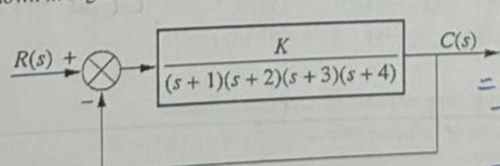
-1

$\pm 1.4j$

CLO # 4: Analyze a given system in frequency domain.

Q4: **[20 marks]**

a) For the system shown in Figure below,



i) $\sigma = \frac{\Sigma p - \Sigma z}{\#p - \#z}$

$= \frac{(-1-2-3-4)-0}{4-0}$

$= -5/2 = -2.5$

i. Find the asymptotes of the root-locus.

ii. Find the break-away and break-in points (if any) of the root-locus.

iii. The crossings on the $j\omega$-axis of the root-locus.

iv. Hence, sketch the root-locus.

$\theta_\sigma = \frac{(2n+1)180}{\#p - \#z}$

$n=1, \frac{((2\times1)+1)180}{4-0}$

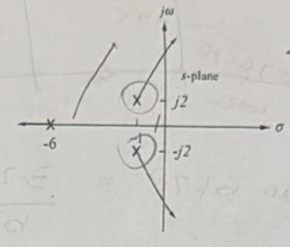
$n=2, \frac{((2\times2)+1)180}{4-0}$

2, 4

b) The root locus of a system, **G(s)**, is shown here:

i. Find the open-loop transfer function, **G(s)**, of the system.

ii. Hence, find the value of associated gain K for which the closed-loop function will yield a closed-loop pole on the real-axis at **s = - 8**.



$$\frac{K}{(s+6)(s+1+j2)(s+1-j2)}$$

$KG(s) = -1$

CLO # 5: Design a closed-loop feedback controller

Q5: The open loop transfer function of an un-compensated unity feedback system is given below. **[20 marks]**

$$KG(s)H(s) = \frac{K}{(s+1)(s+4)}$$

The un-compensated system has a dominant pole at **-2.5 + j1.875**. The damping ratio of the un-compensated system is **0.8**.

i. Find the un-compensated system's settling-time and peak-time.

ii. It is desired to compensate the system to decrease its settling-time by a factor of 2.0 and steady-state error to zero without affecting the damping ratio. Find the new settling-time, peak-time, and the dominant pole location to satisfy the required performance metrics.

iii. Hence, design a PID controller to compensate the system to achieve the performance in part (ii). Write the overall transfer function of the system including plant and controller pole/zeroes and gain.

$$(s+a7)\left(\frac{s+0.01}{s}\right)$$

$$\frac{(s+7)(s+0.01)}{s} = \frac{s^2+0.01s+7s+0.07}{s}$$

$b = 5.778$

$= 2.352$

$\theta_2 \rightarrow \quad \tan\alpha_2 = \frac{3.76}{4} = 43.15 \Rightarrow 180 - 43.15$

$$\boxed{\theta_2 = 136.85^\circ}$$

$\theta_1 : \quad \tan\alpha_1 = \frac{3.75}{1} = 75.07^\circ \Rightarrow 180 - 75.07$

$$\boxed{\theta_1 = 104.93^\circ}$$

$+\theta_z - \theta_1 - \theta_2 = 180$

$\theta_z = 421.78 - 360$

$$\boxed{\theta_z = 61.78^\circ}$$